

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2024/2025

DECEMBER EXAMINATION

UCCD2103 OPERATING SYSTEMS

MONDAY, 23 DECEMBER 2024

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING

Instructions to Candidates :

This question paper consists of THREE (3) questions in Section A and TWO (2) questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Should a candidate answer more than ONE (1) question in section B, marks will only be awarded for the FIRST ONE (1) question in that section in the order the candidate submits the answers.

Candidates are allowed to use calculator.

Answer questions only in the answer booklet provided.

UCCD2103 OPERATING SYSTEMS**Section A: Answer ALL questions.**

- Q1. (a) Draw and label the diagram of the 5-State process model. (5 marks)
- (b) Referring to the following sequential scenarios, define each state transition of **Process A** based on the 7-state process model:
- (i) Process A is created and placed in the ready queue. (1 mark)
 - (ii) Process A then starts executing. (1 mark)
 - (iii) Process A requests data from a disk, causing it to wait. While waiting, the execution is switched to another process, Process B. (1 mark)
 - (iv) The disk operation completes but Process A is not scheduled to execute yet because the CPU is still executing Process B. (1 mark)
 - (v) Process B requests for a large amount of memory causing Process A to be suspended to virtual memory. (1 mark)
 - (vi) Process A is then selected to resume execution until finish. (1 mark)
- (c) Explain the motivation behind the evolution of the Five-State process model to the Seven-State process model. (4 marks)
- (d) Define the following terms:
- (i) Critical section (2 marks)
 - (ii) Deadlock (2 marks)
 - (iii) Race condition (2 marks)
- (e) For each of the following use cases, determine the type of real-time task (hard or soft real-time) in the application or system and justify your answer:
- (i) A system that controls the braking mechanism in an autonomous vehicle must respond instantly to avoid collisions. (2 marks)
 - (ii) Consider a video conferencing application used for virtual meetings. The application needs to process and transmit audio and video data in real-time to ensure smooth communication. Occasional delays or glitches in the audio or video stream may cause a brief lag or a momentary freeze. (2 marks)
- [Total : 25 marks]

UCCD2103 OPERATING SYSTEMS

- Q2. (a) This question is about the Completely Fair Scheduler (CFS) in the Linux kernel. Assume that the system has a uni-processor and the period for CFS is set to 20 milliseconds. There are two CPU-bound processes in Ready state; A and B with the parameters shown in Table 2.1:

Table 2.1: Parameters for Process A and B

Parameters	Process A	Process B
Virtual runtime (nanoseconds)	1,284,317	1,312,086
Total time quantum allocated (milliseconds) per period	18.4119	X
Nice value	Y	4

The minimum virtual runtime of the red-black tree is 1,284,317 nanoseconds. Using the guide provided in the Appendix A, answer the following questions:

- (i) Calculate the values of X and Y in Table 2.1. (6 marks)
 - (ii) In what order will Process A and Process B execute in the next iteration? Briefly provide your reason. (2 marks)
 - (iii) What are the virtual runtimes of Process A and Process B after they execute for one period? (6 marks)
- (b) A process is divided into eight pages on disk and is assigned a fixed allocation of **two** page frames in the main memory. The following page trace occurs:

5,6,7,4,8,4,7,5,6,8,4,5

Assume that the frames are initially empty. Show the successive pages residing in the two page frames using:

- (i) Clock replacement policy. Indicate the Use bit. (6 marks)
- (ii) Least Recently Used replacement policy. (5 marks)

[Total: 25 marks]

UCCD2103 OPERATING SYSTEMS

- Q3. (a) Consider a disk head is now positioned at track 200 and has completed a previous request at track 220. Determine the average of disk head movement for the following requests of track if C-SCAN disk scheduling algorithm is implemented. Show every movement in your answer. (8 marks)

195, 275, 265, 190, 125, 120, 230

- (b) Briefly explain the following terms related to disk performance.
- (i) Seek time (2 marks)
- (ii) Rotational delay (2 marks)
- (c) The following bit table represents Block 0 to Block 23 in a disk, which at the moment contains only File T.

111001001100001100010000

Assume that *Indexed Allocation with Block Portions* was implemented and Table 3.1 shows a portion of the file allocation table.

Table 3.1: File allocation table

File name	Index Block
File T	5
File N	Y

- (i) List the contents of the Index Block for File T. (4 marks)
- (ii) 8 blocks are required to store a new file named File N. Given that the searching for free blocks to store new file will start from the first block, Block 0, and the first available free block will be used as the Index Block. What is the value of Y in Table 3.1? (1 mark)
- (iii) List the contents of the Index Block for File N. (4 marks)
- (d) Consider an application that stores records in a file with the size of each record equals to the size of a block in a disk. The application must be able to retrieve individual record from the file upon request. Briefly discuss the suitability of index allocation versus linked allocation in storing the file in a disk. (4 marks)

[Total: 25 marks]

UCCD2103 OPERATING SYSTEMS**Section B: Answer ANY ONE (1) question.**

- Q4. (a) Table 4.1 shows the current allocation of resources to each process, the processor time consumed and the resources that are still needed by each process to proceed and complete their execution.
 The resources currently available are $R1 = 1$, $R2 = 0$, $R3 = 0$ and $R4 = 0$. All the processes are currently deadlocked.
 Resolve the deadlock by *successively abort deadlocked processes with the least processor time consumed*. Show all the steps and the resources available after each abortion in your workings. (9 marks)

Table 4.1: Resources allocation of processes

Process	Processor time consumed	Allocated				Needed			
		R1	R2	R3	R4	R1	R2	R3	R4
A	22	2	1	1	0	3	2	2	4
B	10	1	0	2	1	3	4	2	5
C	18	2	1	0	1	5	4	4	2
D	14	1	1	2	2	3	2	5	5
E	12	0	2	1	1	4	4	6	5

- (b) Given the size of a user process is 32GB, page frame size is 8KB and each page table entry consumes 4 bytes, calculate the size of the root page table in two-level scheme. Show all your calculations. (4 marks)
- (c) Table 4.2 shows a portion of a segment table for a process and it is known that the segment table can have a maximum of 128 table entries. Translate the following virtual addresses to their corresponding physical addresses. Show all the relevant values in your answer.

Table 4.2: Portion of a segment table

Segment number	Base	Length (decimal)
39	0000 0100 0010 0001	511
40	0000 0010 0110 0100	510
41	0000 0111 0011 0101	508

- (i) 0100111110100111 (3 marks)
- (ii) 0101001111111110 (3 marks)
- (d) Briefly explain how the multiprogramming level relates to
- (i) higher process suspension rate. (3 marks)
- (ii) thrashing problem. (3 marks)

[Total: 25 marks]

UCCD2103 OPERATING SYSTEMS

- Q5. (a) Given that a system has 1MB of memory available, apply the Buddy system allocation to the followings.

NOTE: Answer can also be shown in the form of binary tree.

- (i) Memory requests in the sequence shown in Table 5.1. (4 marks)

Table 5.1: Memory requests

Process	Request
A	100 KB
B	200 KB
C	50 KB
D	120 KB
E	25 KB
F	220 KB

- (ii) The memory partitions allocated to the process C, process D, process E and process F are released. (2 marks)

- (b) Analyse and explain how race condition could occur in the pseudocodes given in Figure 5.1. The global variables are shared by both threads. (5 marks)

Global variable p=2, t=4, k;	
Thread_1(){ k = t + p/2; }	Thread_2(){ k = t * p/2; }

Figure 5.1: Pseudocode of two threads with global variables

- (c) Analyse the pseudocodes given in Figure 5.2 and then describe an execution sequence that could lead to a deadlock situation, and another sequence that would avoid deadlock. Assume any process can be selected to execute first, and process switching can occur at any time during execution. State the changes of the semaphores' value in your explanation. You may use the line number (L1, L2, etc.) in your answer. (7 marks)

semaphore x=1, y=0;	
G()	K()
L1 {	{
L2 semWait(x);	semWait(x);
L3 semWait(y);	//critical section
L4 //critical section	semSignal(y);
L5 semSignal(x);	semSignal(x);
L6 }	}

Figure 5.2: Pseudocode of two processes

UCCD2103 OPERATING SYSTEMS**Q5. (Continued)**

- (d) Figure 5.3 shows the view of a part of a production line, where the **Packager** will pack the product and place it on the platform. The **Labeller** will then label the product and remove it from the platform. The operation of Packager and Labeller can be simulated by two threads, namely **PK** and **LB** respectively. Figure 5.4 shows the skeletons of threads PK and LB. The **place()** function in thread PK will pack the product and place it on the platform whereas the **remove()** function in thread LB will label the product and remove it from the platform. Packager must wait for the platform to be emptied whereas the Labeller must wait for product to be placed on the platform before they can proceed with their operation.



Figure 5.3: Part of a production line

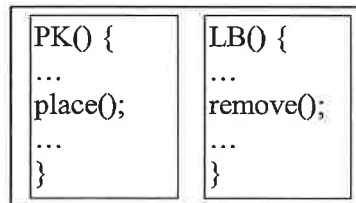


Figure 5.4: Skeletons of thread PK and thread LB

- (i) What is the number of semaphores required to synchronise the execution of the two threads? (1 mark)
- (ii) Assume that the platform is initially empty, write the pseudocode for the two threads in Figure 5.4. Initialise each semaphore with the appropriate value. (6 marks)

[Total: 25 marks]

UCCD2103 OPERATING SYSTEMS

APPENDIX A:

Linux Completely Fair Scheduler (CFS):

1. Tasks' time slice allocation:

$$f_i(t) = \frac{w_i}{\sum w_{rbtree}} \times p$$

$f_i(t)$ = time slice received by task i at time t ,
 w_i = weight of task i

$\sum w_{rbtree}$ = Total weight of tasks in red-black tree

p = period

2. Virtual runtime of task i in n -th iteration:

$$v_i(n) = v_i(n-1) + \left(\frac{\Delta e}{w_i} \right) \times w_{nice0}$$

$v_i(n-1)$ = virtual runtime of task i in previous iteration

Δe = elapsed time task i spent utilizing the processor during iteration n

w_i = weight of task i

w_{nice0} = weight assigned to nice 0 task = 1024

3. Mapping of nice value to weight:

```
static const prio_to_weight[40] = {
/* -20 */      88761,      71755,      56483,      46273,      36291,
/* -15 */      29154,      23254,      18705,      14949,      11916,
/* -10 */      9548,       7620,       6100,       4904,       3906,
/* -5 */       3121,       2501,       1991,       1586,       1277,
/* 0 */        1024,        820,        655,        526,        423,
/* 5 */         335,        272,        215,        172,        137,
/* 10 */        110,         87,         70,         56,         45,
/* 15 */         36,         29,         23,         18,         15,
}
```